

PLANWREN | FREE SAMPLE

One page from the full book follows.

Sample only. Full book is not included.

Test 1

READING (60 minutes, 40 questions)

Reading Passage 1

You should spend about 20 minutes on Questions 1-13, which are based on Reading Passage 1 below.

Growing Upwards: The Promise and Limits of Vertical Farming

A For most of human history, farming has meant fields: wide, flat stretches of soil exposed to sun, wind and rain. Vertical farming turns this idea on its head, quite literally. Instead of spreading crops across the land, growers stack them in layers, one above another, inside buildings. The shelves may rise several metres high, and the plants on them may never see natural daylight. Supporters describe it as a way to grow fresh food in the heart of a city. Critics say it is an expensive solution to problems that ordinary farms already solve quite well.

B The basic method is simple to describe. Plants sit in trays on racks, and instead of soil their roots are fed by water that carries dissolved nutrients. In some systems, known as hydroponics, the roots hang in a slow stream of this water. In others, called aeroponics, the roots hang in the air and are sprayed with a fine mist at regular intervals. Light comes from rows of LED lamps, which can be tuned to the particular colours that plants use most for growth. Temperature, humidity and even the level of carbon dioxide in the air are controlled by computer.

C This level of control brings clear benefits. Because the growing room is sealed, pests find it hard to get in, so many vertical farms use few or no chemical pesticides. Water that is not taken up by the plants can be collected, cleaned and used again, and growers often claim that they need a small fraction of the water used by a field farm to produce the same amount of lettuce. Weather no longer matters either: a vertical farm can harvest the same crop every few weeks throughout the year, whether there is a drought, a flood or a frost outside.

D Location is another selling point. Most of the fresh vegetables eaten in large cities travel long distances before they reach a shop, and some of them spoil on the way. A farm on the edge of a city, or even inside an old warehouse near the centre, can deliver salad leaves to supermarkets within hours of picking. Shoppers get fresher food, and less is thrown away. Some projects have been built in places where farming land is scarce or the climate is harsh, such as desert cities and small island states, where most vegetables would otherwise have to be imported.

E However, there is one resource that vertical farms use far more of than field farms: electricity. The sun provides light to an outdoor farm for free, but indoors every hour of light must be paid for. Lighting, together with the cooling needed to remove the heat that the lamps produce, makes up a large share of a vertical farm's running costs. When prices for electricity rose sharply in several countries in the early 2020s, a number of well-funded vertical farming companies ran into financial trouble, and some closed altogether. The environmental case also depends heavily on where the power comes from. A farm run on electricity from coal may be responsible for more carbon emissions than the lorries it replaces.

F The range of crops is limited, too. Leafy greens such as lettuce, spinach and basil grow quickly, stay small and can be sold at a good price, which makes them ideal for stacked shelves. Staple crops are a different matter. Wheat, rice and maize need large amounts of light over a long growing season, and each plant produces only a small amount of food compared with its size. Growing them indoors would cost many times more than growing them in a field, so for now these crops remain firmly outdoors. Some farms are experimenting with strawberries, tomatoes and peppers, but these